Lab 03 Instruction EE160

Single-Phase AC Circuit Measurements

Resistive and Series RL Loads at 60 Hz

1. Learning Objectives

1. Measure voltage and current in single-phase AC hardware circuits.
2. Compare resistive-load measurements with theoretical calculations.
3. Measure the change in voltage–current phase angle after adding a series inductor.
4. Relate phase angle to real, reactive, and apparent power and power factor.
5. Explain the limitations of phase measurements for a nominally resistive load.

2. Background Theory

2.1 Resistive and Series RL Loads

For an ideal resistor, voltage and current are in phase. All voltage and current values below are RMS unless stated otherwise.

$$I = V/R \qquad P = I^2R = V^2/R \qquad \varphi = 0°$$ For a

sinusoidal waveform, $V_{rms} = V_{pk}/\sqrt{2} = V_{pp}/(2\sqrt{2})$. For a series RL load:

$$X_l = 2\pi fL \qquad Z = R + jX_l \qquad |Z| = \sqrt{(R^2 + X_l^2)}$$

$$I = V/|Z| \qquad \varphi = tan^{-1}(X_l/R)$$ A

positive φ means current lags voltage.

2.2 Power and Power Factor

$$S = VI\ (VA) \qquad P = VI\ cos\ \varphi\ (W)$$

$$Q = VI\ sin\ \varphi\ (var) \qquad PF = P/S = cos\ \varphi$$

For the RL circuit, V is measured across the complete resistor–inductor combination.

3. Equipment and Circuit Preparation

Equipment: GFCI, student-built fused receptacle assembly, 120 V-to-12 V isolation transformer, oscilloscope, voltage probe(*remark: preferably to use a differential voltage probe; as we now have a transformer, we can use a normal voltage probe, provided we do not confuse the ground with other channels) , AC-current multimeter, current probe, power resistors, inductor, and insulated leads.

3.1 Supply Arrangement

120 V mains → GFCI → power plug → isolation transformer → AC load. A 0.2 A fuse is connected in the 120 V input hot conductor.

Parameter	Value
Nominal secondary voltage	12 V
Frequency	60 Hz
Primary hot-conductor fuse	0.2 A

Practice 1 resistor	____19.99__ Ω (at least 20 Ω)
Parameter	Value
Practice 2 resistor	____19.99__ Ω
Inductor	___1.25___H

3.2 Before Energizing

1. Have the TA inspect the mains assembly and circuit. Keep mains connections enclosed. Perform all load measurements on the 12 V secondary side.
2. Disconnect power before changing connections. Verify component, transformer, lead, and instrument ratings.
3. Connect the voltage probe across the load. In Practice 1, connect the multimeter in series using the correct AC-current terminals and range. Never place a current-configured meter across the supply.
4. Keep the oscilloscope protective earth connected. A GFCI does not make exposed mains wiring safe.

Don't touch any exposed connection point during measurement.

5. After measurements, disconnect power and allow resistors to cool.

The 0.2 A fuse is on the primary side; it is not a secondary-current limit. Record actual secondary voltage under load rather than assuming it is exactly 12 V.

4. Practice 1 Resistive Load

A. Theoretical Calculations

Use R_1 = ___19.99___ Ω, rated ___100___ W. With power disconnected, measure the actual resistance. Using 12 V RMS, calculate RMS current, peak voltage, and resistor dissipation.

$$I = 12/R_1 = 0.6/\sqrt{2} = 0.424 \text{ A} \qquad V_{pk} = 12\sqrt{2} = 16.97 \text{ V} \qquad P = 12^2/R_1 = 7.203 \text{ W}$$

B. Measurement Procedure

1. With power disconnected, connect the transformer secondary, AC-current multimeter, and R_1 in a series loop. Connect the voltage probe across R_1.
2. Verify probe attenuation and meter settings. Obtain TA approval, then energize the circuit.
3. Record voltage frequency, peak voltage, and RMS voltage on the oscilloscope. Record RMS current on the multimeter. Save the voltage waveform. (You may press the measurement button on the oscilloscope to get the peak voltage, frequency and RMS voltage)
4. Compare $V_{pk}/\sqrt{2}$ with the measured RMS voltage. Recalculate expected current using measured resistor voltage and measured resistance: $I = V_{rms}/R_1$.
5. Complete the table, then disconnect power.

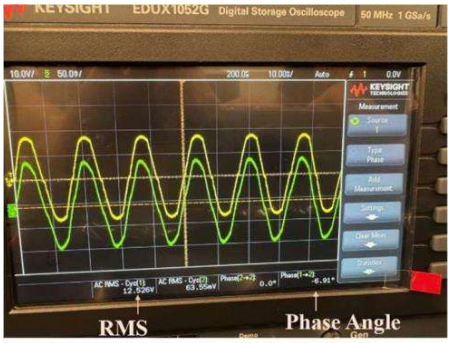


Fig. 1 Example of the measurement

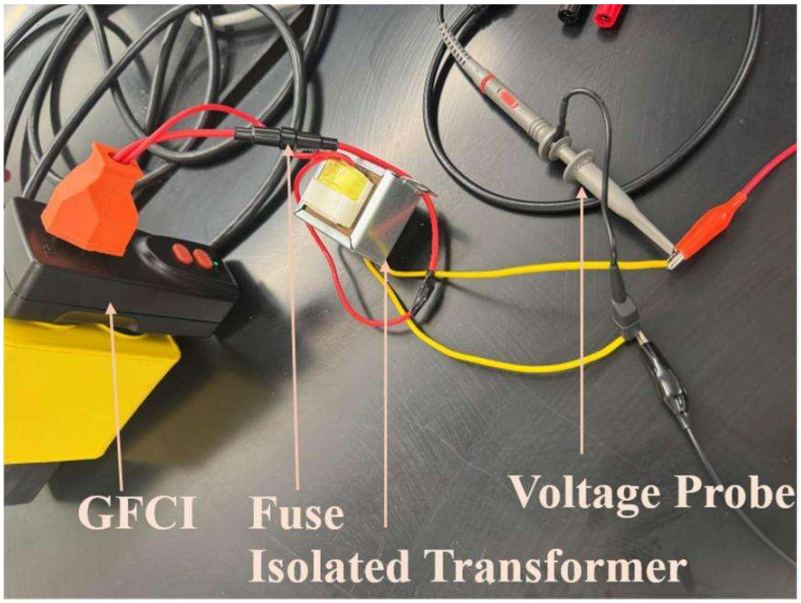


Fig. 2 Example of the voltage probe connection

Measured R_1: _19.99_____ Ω Frequency: _60_____ Hz Voltage peak: ___31.0___ V

Parameter	Theory at 12 V	Experimental	Error (%)
Voltage RMS (V)	12	11.2	6.67%
Current RMS (A)	0.424	0.388	8.49 %
Resistor power (W)	7.203	6.275	12.88 %

Estimate experimental resistor power as $V_{rms}I_{rms}$, assuming an approximately resistive load. This is a derived estimate, not a direct power measurement.

$Error\ (\%) = |Experimental - Theoretical| \,/\, |Theoretical| \times 100\%$

C. Questions

1. Does measured current agree with Ohm's law? Compare predictions based on nominal and measured voltage.

Based on Ohm's law the RMS current is 0.395 A, so the measured is not exactly the same. However, the measured can still be considered as agreeing with Ohm's law because the two values are close to each other which means that differences can be due to the multimeter used and any inconsistencies in the voltage and current of the circuit as it runs.

2. Why can the secondary voltage differ from 12 V?

The secondary voltage can vary from 12v due to fluctuations in the primary side voltage being used. The source my not being exactly 120v other efficiencies in the transformer my also cause loss in voltage. Also, if there is a difference in the turn ratio then the voltages can differ from each other.

3. Can the multimeter current reading alone determine phase difference? Explain.

The multimeter cannot determine phase difference because it just gives a value. It doesn't show a waveform, so it cannot be used to determine the phase difference between the voltage and current.

4. Practice 2 Series RL Load

A. Load Settings and Theory

Use R2 = 19.99 Ω, rated 100 W and L = 1.25 H.
Use R_2 =___Ω, rated _____ W, and L ≈___mH.
R_2 for both measurements below. Use RLC meter to measure the inductance. Use this same

Calculate X_l, impedance magnitude, current, phase angle, P, Q, S, and PF. State whether winding resistance is included. Use nominal 12 V for initial predictions and measured load voltage for the final comparison.

Xl = 2πfL = 471.24 Ω, |Z| = $\sqrt{R\text{^}2} + X\text{^}2$ = 471.66 Ω, I = V/|Z| = 25.44 mA, φ = arctan(XL/R2) = 87.57°,

S = VI = 0.3053 VA, pf = cos(87.57) = 0.0424, P = Scos(φ) = 0.0129W, Q = Ssin(φ) = 0.305 var. Winding resistance is not included.

B. Resistive Baseline

1. With power disconnected, connect R_2 across the secondary. Connect the voltage probe across R_2 and install the current probe.
2. Orient the current probe so positive current enters the positive voltage-reference terminal. For a clamp probe, enclose only one conductor. Verify scale factors and complete required probe zeroing or calibration.
3. Energize the circuit after TA's inspection. Display voltage and current together. Record RMS values, frequency, and corresponding phase angle between voltage and current. Save the waveform.
4. Disconnect power.

Positive phase angle: current lags voltage. Negative angle: current leads voltage.

C. Add the Series Inductor

1. With power disconnected, insert L in series with the same R_2. Place the voltage probe across the complete R_2–L load.
2. Keep the same probe orientation, channel assignments, coupling, ranges, and phase-related settings wherever possible. Record any changes.
3. Energize the circuit. Record RMS values, frequency, and phase angle between voltage and current. Save the waveform and disconnect power.
4. Calculate the phase change and compare it with theory.

A positive change means current shifted toward lagging voltage. If the baseline is approximately resistive and the measurement offset is stable, $\Delta\varphi$ approximates the theoretical RL phase angle. Offsets that change with signal amplitude or settings do not necessarily cancel.

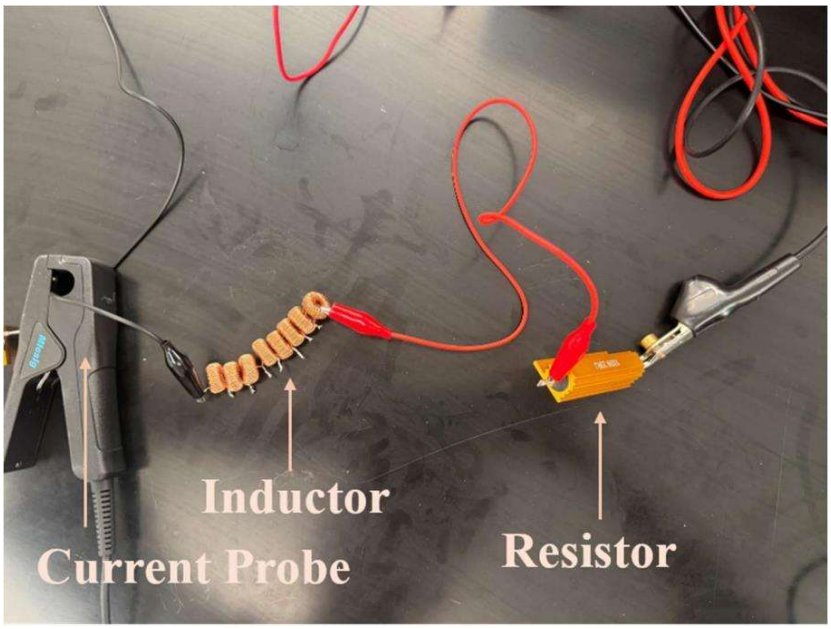


Fig. 3 Example of the current probe connection

5. Practice 2 Results and Analysis

Quantity	R-only baseline	Series RL
Load voltage RMS (V)	11.335V	13V
Current RMS (A)	629 mA	203.025 mA
Raw phase angle (°)	1.17	25.25

Theoretical RL angle: 87.57 ° Measured phase angle: 25.25°

Absolute angular difference from theory: 62.32°

Use absolute angular difference for small angles. Distinguish agreement in direction from quantitative agreement with theory.

If the baseline is stable and the assumptions above are reasonable, use the RL RMS measurements and Δφ to estimate power quantities:

$$S = VI \quad PF = \cos(\Delta\varphi)$$

$$P = S\cos(\Delta\varphi) \quad Q = S\sin(\Delta\varphi)$$

Parameter	Theory	Experimental estimate
Apparent power S (VA)	0.305	2.639
Real power P (W)	0.0129	2.386
Reactive power Q (var)	0.305	1.125
Power factor PF	0.0424	0.904

6. Discussion

1. Why might the resistive baseline show a nonzero phase angle? Why is it important to keep the same resistor, probes, and settings for the comparison?

Resistive baseline might be nonzero due to equipment introducing a phase angle which is why it is important to keep the same probes and settings as changing them could alter the phase angle injected into the circuit. Additionally, the wire rounding inside of resistors can act as inductors at certain frequencies causing a phase angle to be injected and switching resistors could also alter the phase angle being injected which is a reason the same resistor should be used.

2. Did adding L shift current toward lagging voltage? Explain how both raw readings could still show current leading voltage.

Adding L shifted the current toward lagging voltage for the theoretical, but that wasn't the case for the experiment. Both raw readings could still show current leading voltage because the polarity for the voltage probes with connection to the oscilloscope could be inverting the waveform resulting in a large phase shift. Also, it could be due to the load not being large enough as the winding capacitance of the transformer could be larger than the inductance causing the current to lead the voltage. In this case the inductor is much larger than anything that the transformer capacitance could be meaning that it showing leading instead of lagging is must likely due to errors being made with the equipment that was used.

3. How well does the measured change agree with theory? Discuss winding resistance, signal noise, zerocrossing placement, and baseline stability.

The measured change does not agree with the theory at all. This could be due to the inductor that was used causing instability when running the circuit with a small frequency. Also, the transformer has its own winding resistance which isn't taken into account when calculating the theoretical values. Finally, there could be some errors in the collection of the data due to some noise being present and that the values were very low making it difficult to measure them.

7. Report Requirements

Submit circuit sketches showing instrument connections, component values and ratings, theoretical calculations, completed tables, waveform captures, signed phase calculations, discussion answers, and a brief conclusion.

Conclusion:

The experimental and measured values conformed in part 1 of this lab while part 2 the experimental and measured values diverged. We believe that this is most likely due to improper use of the current sensor as that is where our measurements started to significantly diverge from the theoretical.